High-Density Pulsed Electromagnetic Power Generator (HDEP-PG)

Technical Design Specifications

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1 System Architecture

The system utilizes a central pulse-driven magnetic core (20cm) surrounded by a 49-coil independent pick-up array. The electromagnetic architecture integrates inductive coupling with thermoelectric recovery.

2 Input/Output Matrix

The following table defines the operational parameters of the system.

Table 1: Operational Parameters

Parameter	**Value**	**Unit**
Input Voltage (V_{in})	24	Volts (DC)
Input Peak Current (I_{in})	20	Amperes
Switching Frequency (f)	1	kHz
Magnetic Core Length	20	cm
Series Array (29 Coils) Output	High-Voltage	V_{out_HV}
Parallel Array (20 Coils) Output	High-Current	I_{out_HC}
Secondary Energy Recovery	Thermo-Electric	P_{TEG}

3 Governing Physics

3.1 Inductive Kickback (Back-EMF)

The core mechanism is defined by Faraday's Law of Induction, where the sudden current break induces a high-voltage spike:

$$\varepsilon = -L\frac{di}{dt} \tag{1}$$

Where:

- ε is the induced electromotive force.
- L is the self-inductance of the primary coil.
- $\frac{di}{dt}$ represents the rapid decay of current during the switching break.

3.2 Thermal Energy Harvesting

Waste thermal energy from the SiC casing is converted via the Seebeck Effect:

$$P_{TEG} = \alpha \cdot \Delta T \cdot I_{TEG} \tag{2}$$

Where α is the Seebeck coefficient and ΔT is the temperature gradient maintained by the ceramic insulation.

4 System Constraints

- **Core Material:** Laminated iron (3mm thickness) to mitigate eddy currents at 1kHz.
- **Dielectric Housing:** Silicon Nitride (Si_3N_4) for thermal stability and high-voltage insulation.
- **Thermal Interface:** Silicon Carbide (SiC) coupled with thermoelectric modules for hybrid energy harvesting.